

# Ahsanullah University of Science and Technology

## Department of CSE

Fall 25

Course Name: Formal Language and Compilers Lab

Course No: CSE 4130

### Assignment 1

Write a C/C++ program that reads a C source file containing single-line and multi-line comments, along with white spaces (**extra spaces, tabs, and newline characters**). As the first step toward compilation, your program should remove all comments and unnecessary white spaces without removing the spaces required for correct syntax, and write the filtered code to a new output file. The program must also display the contents of both the input and the output files on the console.

#### Sample Input: Input.c

```
1  #include<stdio.h>
2
3  int    main(void)
4  {
5
6      // Single Line Comment
7      printf ("Hello");
8      /* Multi
9      Line
10
11          Comment
12
13      */
14      printf("World");
15      return 0;
16  }
17
```

#### Sample Output:

```
Group_20210204XYZ.ex x + v
Sample Input: Input.c
#include<stdio.h>

int    main(void)
{

// Single Line Comment
printf ("Hello");
/* Multi
Line

        Comment

    */
printf("World");
return 0;
}
Sample Output: output.txt
#include<stdio.h> int main(void) { printf ("Hello"); printf("World"); return 0; }
Process returned 0 (0x0)    execution time : 0.636 s
Press any key to continue.
```